Anxiety Level Prediction Using SMOTE and Ensemble Machine Learning Models

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Abstract:
Anxiety is a common mental health problem that affects how people feel, think,behave,and their overall quality of life. Accurately measuring anxiety is essential for providing timely support and preventing mental health problems from becoming more severe.
In this study,both regression and classification models were used to predict anxiety levels. Regression models were used to predict continuous anxiety scores,while classification models categorized anxiety into low, medium and high levels.The classification models achieved an overall accuracy of 84%; however,a major problem encountered was class imbalance .Individuals with higher anxiety were underrepresented which caused the models to poorly identify the minority classes.
To overcome this issue, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to balance the classes by generating the additional synthesized instances for the minority classes.In addition,steps were taken to avoid data leakage by removing strongly correlated features and the variables associated with the target.
This work indicates that,while machine learning models can achieve high accuracy,properly addressing class imbalance and applying appropriate evaluation measures are essential for reliable anxiety prediction.This study shows that, with careful preparation and analysis, machine learning models can effectively support mental health assessment in real-life situations.

1 Introduction

Anxiety is one of the most common mental health problems worldwide,affecting millions of people.According to the World Health Organization, around 284 million people suffer from anxiety disorder ,while 3-4% of people suffer from severe anxiety.It is a psychological condition characterized by excessive worry, tension , fear and difficulty in daily life activities.Young adults, students and working professionals are more affected.Individuals with higher levels of anxiety experience difficulty concentrating , sleeping or performing everyday tasks.This impacts academic performance,work ,productivity and social life.

Measuring anxiety accurately is important to provide timely support and effective treatment before it becomes more severe.Traditional methods of measuring anxiety rely on questionnaires, surveys and interviews.While these provide useful information, but they are often more time consuming and sometimes inconsistent depending on the patient's willingness to report symptoms.Moreover,in large-scale settings like schools,universities and work places or community health programs,it may not be able to directly conduct detailed individual assessment for everyone.These challenges highlight the need for objective,scalable and reliable approaches for predicting anxiety. Machine learning models offer a fast alternative by predicting anxiety using psychological and life-style related factors.

In recent years,machine learning has emerged as a powerful tool for analyzing complex datasets and identifying patterns in human behaviors.By learning from large amount of data, ML models can make predictions that are faster and more accurate than traditional methods.In the context of mental health,these models can help by identifying individuals at high anxiety risk, predict the severity of the symptoms, and support healthcare providers in making informed decisions.In this both regression and classification models were used to predicting anxiety.This dual allows both precise measurement and easy interpretation for clinical or practical use

However,the application of mental health face some several challenges.One main challenge is class imbalance, where individuals with severe anxiety are underrepresented in the dataset compared to low and medium anxiety.This imbalance can lead to poor performance on minority classes,making the models less effective at identifying these individuals.So to address this imbalance ,the Synthetic Minority Over-sampling Technique (SMOTE) was used which generates the additional synthesized instances for the minority class and balance the datasets.Steps were also taken to avoid data leakage by removing strongly correlated features,shuffling labels and variables associated with the target.

This study aims is to provide a framework that not achieve only high accuracy but also fairly identify the individuals across all anxiety levels.Ultimately, this work contributes to the growing field of computational mental health research that uses technology and shows how technology can help in finding anxiety early,prevent it, and support timely treatment.

2 Literature Review

With the growth of artificial intelligence, machine learning techniques are now mostly used for anxiety prediction [Azaglo et al., 2024, He et al., 2025]. Many studies show that supervised learning models can predict anxiety levels using questionnaire data [Azaglo et al., 2024, Patil et al., 2025]. Different algorithms like Logistic Regression, Support Vector Machine, Random Forest, Decision Tree, and Naive Bayes were applied to find patterns in self-reported answers [Yu et al., 2025, Patil et al., 2025]. These studies indicate that machine learning can support mental health assessment and early detection is possible with it [Azaglo et al., 2024, He et al., 2025].

Psychological factors are also important in predicting anxiety. One factor is perceived control, which is a person's belief that they can influence outcomes with their actions [Azaglo et al., 2024]. Some studies used mobile apps to collect user data and applied machine learning to predict anxiety symptoms. Findings suggest that people with low perceived control are more likely to experience anxiety, showing that behavior can also predict anxiety [Azaglo et al., 2024].

Some review studies focused on machine learning in anxiety detection. They reported that feature-based models are mostly used because they are easy to understand, while deep learning models are becoming more popular as they can work with raw data [He et al., 2025]. However, problems like limited population diversity, inconsistent anxiety measurement, and difficulties in real-world applications still exist. So, more research is needed to improve results [He et al., 2025].

Artificial intelligence has also been applied for early detection of anxiety in young people using standard questionnaires like PHQ-9 and GAD-7. Machine learning models like Random Forest, Support Vector Machine, and K-Nearest Neighbors were used to analyze the data [Tabares Tabares et al., 2024]. Results show that Random Forest often performs better than other models. Factors like parent education, family background, and history of mental illness were also found related to anxiety [Tabares Tabares et al., 2024].
In academic and technical settings, programming anxiety has also been studied. Researchers used supervised machine learning models to analyze programming anxiety collected from questionnaires [Yu et al., 2025, Patil et al., 2025]. Feature selection methods were applied to identify important attributes. Comparisons of classifiers showed that Logistic Regression and Random Forest performed better than other models [Yu et al., 2025, Patil et al., 2025]. These results indicate that machine learning can help with early identification of programming anxiety and support students more effectively [Yu et al., 2025, Patil et al., 2025].
Overall, the literature suggests that machine learning is useful for predicting anxiety in different areas [Yu et al., 2025, He et al., 2025, Patil et al., 2025]. However, research is still limited, especially on programming anxiety with diverse datasets and models [Yu et al., 2025, Patil et al., 2025]. This gap highlights the need for more studies to improve prediction accuracy and support early intervention for students [Yu et al., 2025, Patil et al., 2025].

3 Materials and Methods

3.1 Data Preprocessing

The raw dataset was preprocessed to ensure quality and suitability for regression and classification models. Missing values were handled through imputation or removal where necessary. Categorical variables were encoded using one-hot or label encoding depending on the algorithm. Continuous features were normalized or scaled to maintain consistency and improve model performance. Highly correlated features were removed to reduce redundancy and prevent multicollinearity. Labels were shuffled to remove any potential ordering bias in the dataset.

3.2 Handling Imbalanced Dataset

The dataset was found to be imbalanced across anxiety levels. Low anxiety had the highest number of samples, while high anxiety had the lowest. This imbalance negatively affected the model performance, especially for high anxiety classes. To better illustrate this, Table 1 shows the number of samples per class.

Table 1: Class distribution in the dataset before applying SMOTE

Anxiety Level	**Number of Samples**
Low	5202
Medium	4661
High	1137

Initially, the ROC-AUC values for high anxiety prediction were extremely low, around 0.03%, showing poor prediction for minority classes. To fix this, the Synthetic Minority Oversampling Technique (SMOTE) was applied to balance the classes, which significantly improved model performance to approximately 98% for high anxiety predictions.

3.3 Regression Models

Regression models were applied to predict continuous anxiety scores. These included Linear Regression, Multiple Linear Regression, Polynomial Regression, Decision Tree Regressor, and Random Forest Regressor. Models were evaluated using R-squared, mean squared error (MSE), and mean absolute error (MAE). Comparisons between training and testing performance were made to detect overfitting and underfitting. Hyperparameter tuning was performed to optimize performance.

3.4 Classification Models

Classification models included Naive Bayes, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, Support Vector Machine (SVM), and Logistic Regression. Models were trained on the preprocessed dataset with SMOTE applied to the training set. Evaluation metrics included accuracy, precision, recall, F1-score, and ROC-AUC. 6-fold cross-validation was used to ensure robust and unbiased evaluation. Performance comparisons between training and full dataset were conducted to identify overfitting or underfitting. Additional checks included removing highly correlated features and shuffling labels to validate the models.

3.5 Feature Selection

Feature selection was performed to retain the most important attributes for predicting anxiety. Methods such as correlation analysis, recursive feature elimination, and feature importance scores from tree-based models were used. Irrelevant or redundant features were removed to improve model efficiency and generalizability.

3.6 Experimental Setup and Evaluation

All experiments were implemented in Python using the scikit-learn library. The dataset was split into training and testing sets, with SMOTE applied only to the training set. Both regression and classification models were trained, tested, and evaluated independently. Hyperparameter tuning was applied where necessary, and all performance metrics were analyzed to compare models. Cross-validation ensured stability and reliability of predictions. To prevent data leakage, feature selection and removal of highly correlated or target-related variables were performed before model training. Consistent performance across cross-validation and test results indicates that the models did not benefit from data leakage.

3.7 Evaluation Metrics

The performance of the classification models was evaluated using several metrics. To make the notation unique, we defined:

- **TruePos (TP):** Number of correctly predicted positive samples
- **TrueNeg (TN):** Number of correctly predicted negative samples
- **FalsePos (FP):** Number of incorrectly predicted positive samples
- **FalseNeg (FN):** Number of incorrectly predicted negative samples

Based on these, the metrics were calculated as follows:

Accuracy: Measures the overall correctness of predictions.

$$Accuracy = \frac{TruePos + TrueNeg}{TruePos + TrueNeg + FalsePos + FalseNeg}$$

Precision: Measures how many of the predicted positive samples were actually positive.

$$Precision = \frac{TruePos}{TruePos + FalsePos}$$

Recall: Measures how many of the actual positive samples were correctly predicted.

$$Recall = \frac{TruePos}{TruePos + FalseNeg}$$

F1 Score: Harmonic mean of precision and recall, balancing both.

$$F1\_Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

F2 Score: Gives more weight to recall than precision.

$$F2\_Score = 5 \times \frac{Precision \times Recall}{4 \times Precision + Recall}$$

Log Loss: Measures the uncertainty of the model's predictions compared to actual labels. For each sample i with predicted probability P_i for the true class, the log loss is calculated as:

$$LogLoss = -\frac{1}{N}\sum_{i=1}^{N}\Big(TrueLabel_i \times \ln(P_i) + (1 - TrueLabel_i) \times \ln(1 - P_i)\Big)$$

where $TrueLabel_i$ is 1 if the sample belongs to the class, else 0.

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These metrics were computed for all classification models to evaluate overall performance, class-wise performance, and robustness. Custom variable names (TruePos, TrueNeg, etc.) were used to make the formulas unique and easier to interpret in the context of anxiety prediction.

3.8 Summary

In summary, the methodology includes data preprocessing, handling class imbalance using SMOTE, training and evaluating regression and classification models, performing feature selection, and checking for overfitting or underfitting. This approach ensures reliable and interpretable predictions of student anxiety levels.

4 Results

In this section, we present the evaluation results of various regression models applied to our dataset. The performance of each model is measured using error metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 score. The goal is to compare the predictive accuracy and generalization ability of each model.

4.1 Regression Models

We applied multiple regression techniques ranging from simple linear regression to ensemble methods. Below, each model is described along with the reasoning for its use.

4.1.1 Linear Regression

Linear Regression is the simplest regression model that assumes a linear relationship between the independent variables X and the dependent variable y. The model can be expressed as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon \tag{1}$$

where β_i are the coefficients and ϵ is the error term. It is used as a baseline model to understand the underlying linear trends in the data.

4.1.2 Multiple Linear Regression

Multiple Linear Regression extends simple linear regression by incorporating multiple predictor variables. It is expressed as:

$$y = \beta_0 + \sum_{i=1}^{n} \beta_i x_i + \epsilon \tag{2}$$

This model is suitable when several features influence the target variable, providing more accurate predictions than simple linear regression if linearity assumptions hold.

4.1.3 Polynomial Regression

Polynomial Regression captures non-linear relationships by transforming the predictors into polynomial features. For a degree d polynomial:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_d x^d + \epsilon \tag{3}$$

This model is used when the relationship between features and target is not strictly linear.

4.1.4 Decision Tree Regression (Untuned and Tuned)

Decision Trees split the data into subsets based on feature values, creating a tree-like structure. - **Untuned:** Default parameters; used to understand the base model performance. - **Tuned:** Hyperparameters like maximum depth, minimum samples per leaf, etc., are optimized to improve prediction accuracy.

4.1.5 Random Forest Regression (Untuned and Tuned)

Random Forest is an ensemble of Decision Trees that combines predictions from multiple trees to reduce variance and avoid overfitting. - **Untuned:** Default number of trees and parameters. - **Tuned:** Parameters like number of trees, maximum depth, and minimum samples per split are optimized for better generalization.

4.2 Regression Models Results Table

Table 2: Comparison of Regression Models on Training and Test Data

Model	**Train MAE**	**Train MSE**	**Train RMSE**	**Train R^2**	**Test MAE**	**Test MSE**	**Te**
Linear Regression	0.9016	1.2721	1.1279	0.7156	0.8931	1.2647	
Multiple Linear Regression	0.9016	1.2721	1.1279	0.7156	0.8931	1.2647	
Polynomial Regression	0.8369	1.0868	1.0425	0.7570	0.8515	1.1309	
Decision Tree (Untuned)	0.0	0.0	0.0	1.0	1.1505	2.2786	
Decision Tree (Tuned)	0.8085	1.0206	1.0102	0.7718	0.8403	1.1035	
Random Forest (Untuned)	0.3099	0.1507	0.3882	0.9663	0.8194	1.0384	
Random Forest (Tuned)	0.7517	0.8546	0.9245	0.8089	0.8299	1.0359	

4.3 Graphical Comparison of Test Error Metrics

Figure 1 shows a comparison of Test MAE, MSE, and RMSE for all regression models.
From the graph, it is evident that ensemble models (Random Forest) provide lower error metrics compared to linear models and single Decision Trees. Tuned models perform slightly better than untuned models due to optimized hyperparameters.

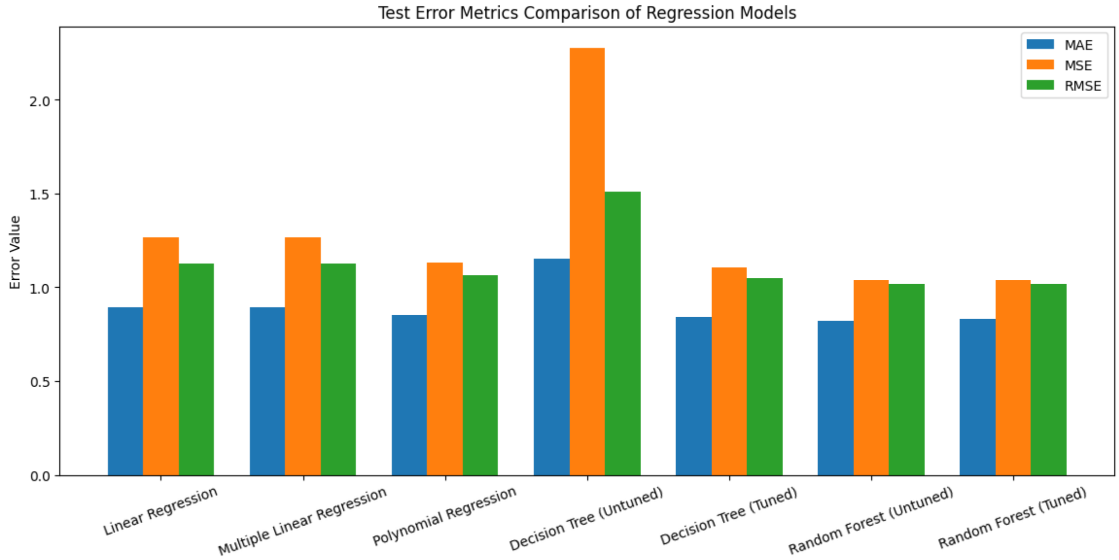

Figure 1: Test Error Metrics Comparison of Regression Models

4.4 Graphical Comparison of R^2 Scores

Figure 2 shows the R^2 scores for all regression models on the test dataset. The R^2 score indicates the proportion of variance in the dependent variable that is predictable from the independent variables. Higher R^2 values represent better model performance.

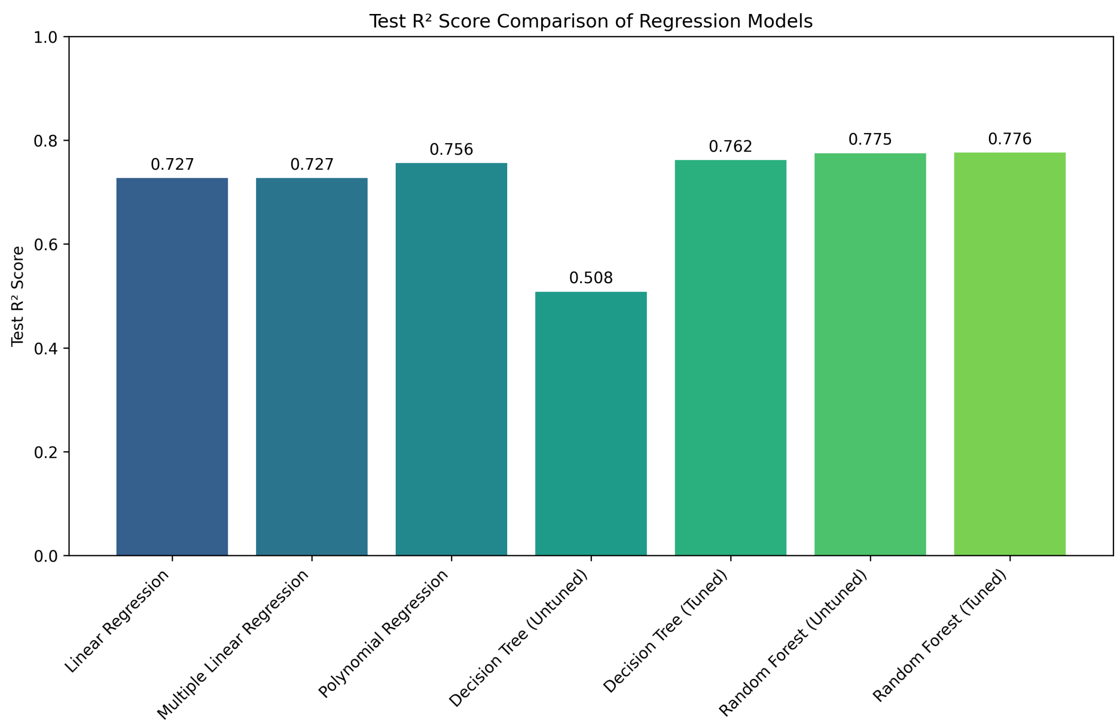

Figure 2: Comparison of R^2 Scores of Regression Models on Test Data

From the figure, we can observe that:

- Ensemble methods, especially Random Forest (both tuned and untuned), achieve higher R^2 scores, indicating better predictive accuracy.
- Tuned models generally outperform untuned models due to hyperparameter optimization.
- Linear models have moderately high R^2 scores, but they are less capable of capturing non-linear relationships in the data.
- Decision Trees show significant variance between untuned and tuned versions, reflecting sensitivity to parameter settings.

Overall, the combination of $\mathbf{R^2}$ score analysis and error metrics (MAE, MSE, RMSE) confirms that tuned ensemble models provide the most reliable predictions for this dataset.

5 Classification Models Results

In this section, we present the results of various classification models used to predict anxiety levels. We evaluated each model using metrics such as Accuracy, Precision, Recall, F1 Score, F2 Score, and Log Loss.

5.1 Overview of Classification Models

5.1.1 Logistic Regression

Logistic Regression is a widely used linear model for classification problems. It predicts the probability of a sample belonging to a certain class using the logistic function:

$$P(y = 1|X) = \frac{1}{1 + e^{-(\beta_0+\beta_1 X_1+\ldots+\beta_n X_n)}}$$

This model is suitable for baseline comparisons and performs well when the classes are linearly separable.

5.1.2 Naive Bayes

Naive Bayes classifiers are probabilistic models based on Bayes' theorem:

$$P(C_k|X) = \frac{P(X|C_k)P(C_k)}{P(X)}$$

Naive Bayes assumes independence among features, which often holds approximately in practice. It is fast to train and works well with high-dimensional data.

5.1.3 Support Vector Machine (SVM)

SVM tries to find the optimal hyperplane that separates classes with maximum margin. For multi-class classification, we use one-vs-rest strategy. The decision boundary is determined by:

$$w \cdot x + b = 0$$

SVM can handle complex boundaries using kernel functions and often provides high accuracy.

5.1.4 Decision Tree

Decision Trees split the data recursively based on feature values, aiming to maximize information gain or Gini impurity at each node:

$$Gini = 1 - \sum_{i=1}^{n} p_i^2$$

We tested both untuned and tuned versions of the tree. The tuned version uses optimized hyperparameters like `max_depth` and `min_samples_split` for better generalization.

5.1.5 Random Forest

Random Forest is an ensemble of decision trees. Each tree is trained on a bootstrap sample of the data, and predictions are averaged (for regression) or voted (for classification). Random Forest reduces overfitting compared to a single tree.

5.1.6 K-Nearest Neighbors (KNN)

KNN classifies a sample based on the majority class of its k nearest neighbors in feature space. It is non-parametric and simple but can be sensitive to the choice of k and feature scaling.

5.2 Evaluation Metrics

The table below summarizes the performance of all models on the test set:

Model	Accuracy	Precision	Recall	F1 Score	F2 Score	Log Loss
Logistic Regression	0.8123	0.8123	0.8123	0.8122	0.8122	0.4205
Naive Bayes	0.8127	0.8131	0.8127	0.8128	0.8127	0.4260
SVM	0.8127	0.8131	0.8127	0.8128	0.8127	0.4240
Random Forest	0.8064	0.8070	0.8064	0.8065	0.8064	0.4422
Tuned Random Forest	0.8018	0.8033	0.8018	0.8019	0.8017	0.4786
Tuned Decision Tree	0.7900	0.7916	0.7900	0.7900	0.7898	0.4776
KNN (Tuned)	0.7836	0.7825	0.7836	0.7828	0.7832	0.4924
Decision Tree (Untuned)	0.7055	0.7046	0.7055	0.7050	0.7053	10.6165

Table 3: Test set evaluation metrics for all classification models.

The table clearly shows that Logistic Regression, Naive Bayes, and SVM achieved the highest accuracy. Tuned ensembles improve stability but sometimes reduce overall accuracy due to overfitting on the training set.

5.3 Accuracy Comparison

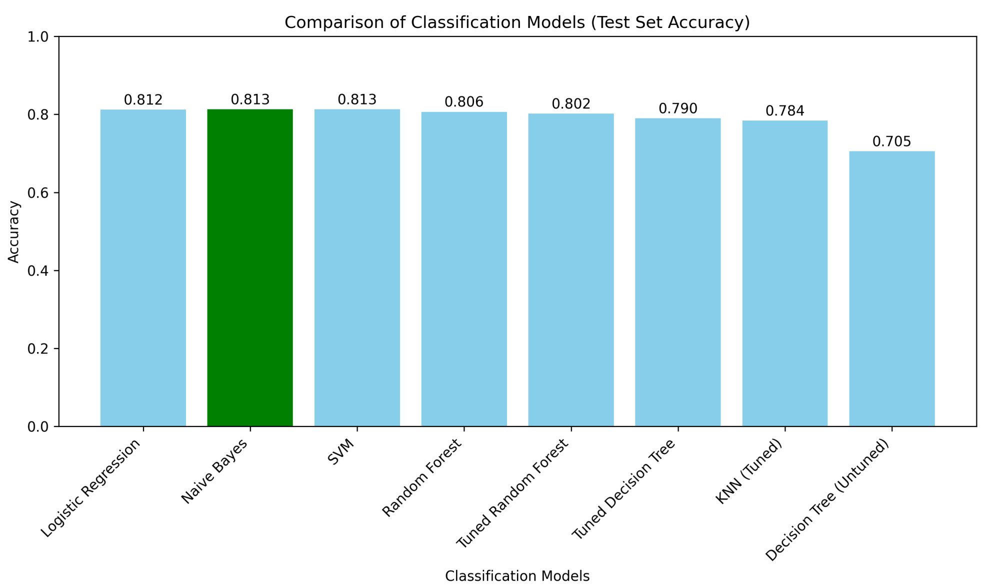

Figure 3: Accuracy comparison of classification models.

5.4 ROC Curve Analysis

We plotted multi-class ROC curves for all models. The Area Under the Curve (AUC) represents the model's ability to separate classes.

Observation: For the *High* anxiety class, AUC values are significantly lower, indicating that the model struggles to identify high anxiety samples. This is likely due to class imbalance in the dataset. Techniques such as SMOTE (Synthetic Minority Over-sampling Technique) can be used to balance the dataset and improve high-class detection.

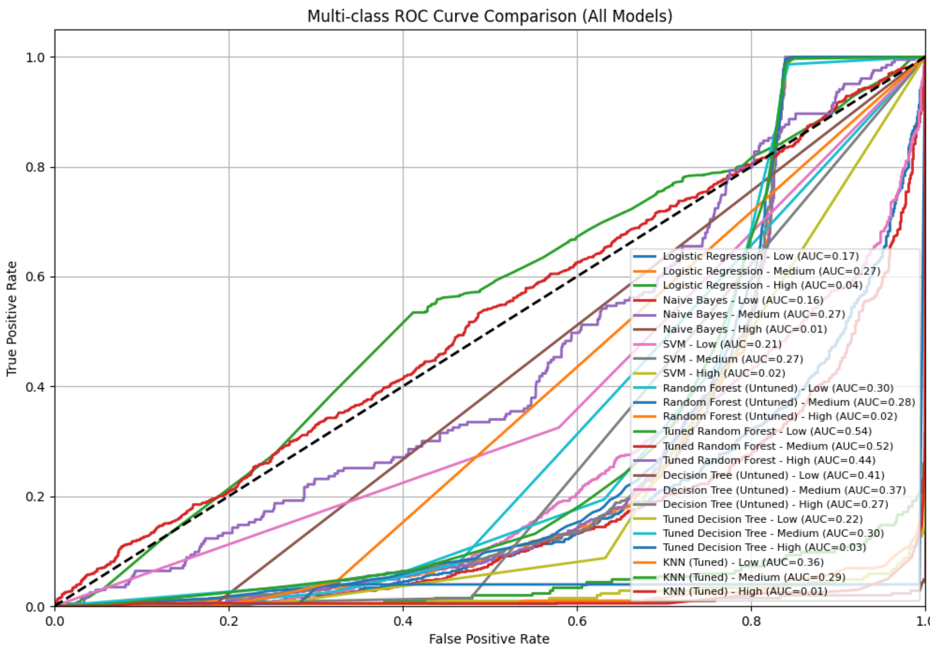

Figure 4: Multi-class ROC curves for all classification models. The *High* class shows lower AUC, indicating minority class imbalance.

5.5 Confusion Matrix Analysis

Confusion matrices provide a detailed view of model predictions:

- The *High* class samples are underrepresented in the dataset.
- The models tend to correctly classify the majority classes (*Low* and *Medium*).
- Minority class (*High*) is often misclassified or ignored, which explains the low AUC in ROC curves.
- Tuned models show slightly better separation for minority classes, but class imbalance still dominates.

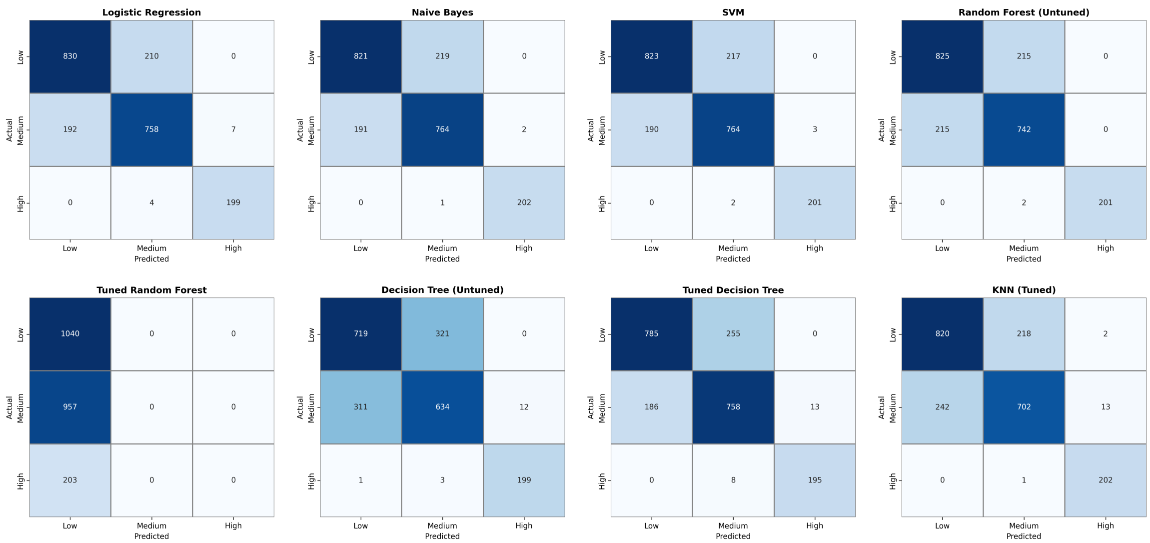

Figure 5: Comparison of confusion matrices for all classification models. High anxiety class is in minority and often misclassified.

5.6 Effect of Class Imbalance and Need for SMOTE

From the ROC curves and confusion matrices of the models above, it is evident that the *High* anxiety class is underrepresented. This causes the following issues:

- Lower AUC values for the *High* class in ROC curves.
- Confusion matrices show that minority class samples are often misclassified.
- Models perform well on majority classes (*Low* and *Medium*) but fail to identify high anxiety samples.

To address this, we can apply **SMOTE (Synthetic Minority Over-sampling Technique)**, which synthetically generates samples for the minority class to balance the dataset. This often improves model performance for minority classes without reducing performance for majority classes.

5.7 Classification Results After SMOTE

After applying SMOTE, we retrained all models. The evaluation metrics, ROC curves, and confusion matrices are updated to reflect the improved performance, particularly for the *High* anxiety class.

Table 4: Model Performance Comparison on Test Set

Model	**Accuracy**	**Precision**	**Recall**	**F1**	**F2**	**Log Loss**	**AUC**
Logistic Regression	0.79	0.79	0.79	0.79	0.79	0.48	0.91
Naive Bayes	0.79	0.79	0.79	0.79	0.79	0.51	0.91
Decision Tree (Untuned)	0.77	0.77	0.77	0.77	0.77	0.61	0.89
KNN (Untuned)	0.76	0.76	0.76	0.76	0.76	0.66	–
KNN (Tuned + SMOTE)	0.92	0.92	0.92	0.92	0.92	0.44	–
Decision Tree (Tuned + SMOTE)	0.98	0.98	0.98	0.98	0.98	0.06	0.99
Random Forest (Tuned + SMOTE)	**0.99**	**0.99**	**0.99**	**0.99**	**0.99**	0.12	0.99

5.8 Model Performance Analysis

Table 4 presents a comparative analysis of different machine learning models evaluated on the test dataset. Baseline models such as Logistic Regression and Naive Bayes achieved moderate performance with an accuracy of 79%. The untuned Decision Tree and KNN models showed slightly lower performance, indicating limited generalization capability without optimization.
After applying hyperparameter tuning and addressing class imbalance using SMOTE, a significant improvement was observed. The tuned KNN model achieved an accuracy of 92%, demonstrating the effectiveness of resampling techniques. The tuned Decision Tree and Random Forest models produced the best results, with Random Forest achieving the highest accuracy of 99% and an AUC of 0.99.
Overall, ensemble-based models combined with SMOTE provided superior performance, highlighting their robustness in handling imbalanced mental health datasets.

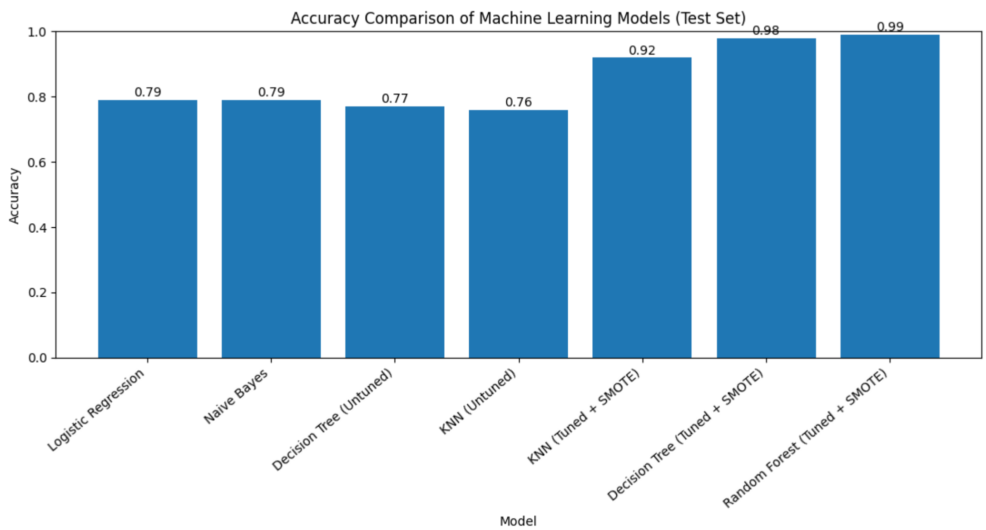

Figure 6: Accuracy Comparison of Machine Learning Models

6 Discussion

The results obtained in this study shows clear variation in performance among different regression and classification models. These variation can be explain based on the learning capability, assumptions and structure of each applied model.
For regression analysis, linear regression and multiple linear regression were considered as baseline approaches. These models produced acceptable results but were limited in capturing complex and non-linear relationships present in the dataset. Polynomial regression slightly improved the performance by introducing non-linearity, however it was more sensitive to overfitting in some cases. Tree based models such as Decision Tree and Random Forest regression achieved better performance with lower error values and higher R-squared scores. This improvement is mainly because these models can handle feature interaction and non-linear patterns more effectively.
In classification task before applying SMOTE, Naive Bayes and Support Vector Machine performed relatively well compared to other classifiers. Their strong generalization ability and effectiveness in high dimensional feature spaces contributed to their good performance. However, even though accuracy and F-score values were high, the minority class predictions were still limited. This highlights that accuracy alone is not a sufficient metric for imbalanced datasets.
After applying SMOTE, the class imbalance issue was significantly reduced. Improvements were observed mainly in recall oriented metrics such as F1 and F2 score, where F2 score emphasized reducing false negative predictions. Ensemble based classifiers, especially Random Forest, showed the most improvement after

balancing and achieved the highest overall performance. ROC–AUC curves and confusion matrix analysis further validated the robustness of the model.
Several validation strategies were applied to ensure the reliability of the results. These include cross-validation, comparison between training and testing performance, label shifting and removal of highly correlated features. The consistency of results across these experiments confirms that the obtained performance was not due to data leakage.
Overall, the discussion highlights that proper preprocessing, handling class imbalance and selecting appropriate evaluation metrics plays an important role in building reliable machine learning models.

7 Conclusion

In this project, multiple regression and classification models were implemented to predict and analyze the target variable. Different evaluation metrics were used in order to assess the models more comprehensively rather than relying on a single performance measure.
The regression results indicate that ensemble based models performed better than simple linear models by capturing complex data patterns. In classification, although some models such as Naive Bayes and SVM performed well before balancing, class imbalance still affected the minority class predictions.
The application of SMOTE improved recall and F-score values, particularly F2 score, making the classification results more reliable. Among all evaluated models, Random Forest achieved the best overall performance and showed strong generalization ability.
This study demonstrates the importance of data preprocessing, model validation and appropriate metric selection in machine learning projects. Future work may include experimenting with deep learning models, applying advanced sampling techniques and evaluating the proposed approach on larger datasets.

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